

Video Transcript - Part 3

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Hello everyone, and welcome to today's session on AWS Storage Essentials. Today, we are unpacking three fundamental storage services in Amazon Web Services: Amazon S3, Amazon EBS, and Amazon EFS.

If you're building or migrating applications in the cloud, one of the first and most critical decisions you'll make is how and where to store your data. Choosing the wrong storage type can lead to poor application performance or unnecessarily high cloud bills. Over the next few minutes, we'll demystify these three services in simple, practical terms so you'll always know exactly which one to pick.

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To understand AWS storage, it helps to step back and look at the core mental models. In AWS, storage isn't 'one size fits all.' Instead, it boils down to two key questions:

1. How is the data accessed? Is your application asking for individual bytes on a raw disk drive, looking up a file path in a folder directory, or fetching an object over HTTP using an API?
2. Who needs access to the data? Is it tied to a single virtual machine, shared across hundreds of servers, or delivered publicly to millions of web users?

AWS built S3, EBS, and EFS specifically to solve these different access patterns. Let's look at each service individually.

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First up is Amazon S3, which stands for Simple Storage Service. S3 is Object Storage. Think of S3 like a giant, virtually bottomless digital locker system. Instead of organizing files into folders and subfolders, S3 stores data as 'objects' inside 'buckets.' Each object contains the file data, a unique key or name, and customizable metadata. What makes S3 special is how you access it: you don't mount S3 to a computer like a hard drive. Instead, you upload and download objects directly using web standards i.e. HTTP and HTTPS APIs. S3 scales practically infinitely and boasts an incredible 11 nines of durability (99.999999999%). This makes S3 the absolute best choice for web images, video streaming, database backups, logs, static web hosting, and massive data lakes.

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Next, let's talk about Amazon EBS, or Elastic Block Store. EBS is Block Storage. If S3 is a digital locker on the web, EBS is like buying an external SSD or hard drive and plugging it directly into your virtual server, your Amazon EC2 instance. Because EBS talks directly to the server at the block level, it delivers extremely low latency, often less than a single millisecond. The server sees an EBS volume as a raw disk that you can format with a filesystem like NTFS or EXT4. However, EBS has a key constraint: by default, an EBS volume attaches to one single

EC2 instance inside a specific Availability Zone. That makes EBS ideal for operating system boot drives and high-performance relational databases like MySQL, PostgreSQL, or SQL Server where ultra-fast read/write speeds are required.

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Our third service is Amazon EFS, or Elastic File System. EFS is File Storage. Think of EFS as a traditional network shared folder, like a network drive in an office where multiple people can open and edit files simultaneously. EFS uses the standard NFS (Network File System) protocol and follows POSIX rules for file permissions. Unlike EBS, which attaches to just one server, EFS can be mounted concurrently by hundreds or even thousands of EC2 instances and containers across multiple Availability Zones. It's also fully elastic: you don't need to pre-allocate storage capacity. If you add 10 gigabytes today and delete 5 gigabytes tomorrow, EFS automatically grows and shrinks, and you only pay for what you actually store. It's ideal for content management systems like WordPress, shared code repositories, and container volumes.

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Let's compare all three side by side to summarize:

1. Storage Type: S3 is Object, EBS is Block and EFS is File.
2. Access Pattern: S3 uses HTTP REST APIs over the internet, EBS acts as a local disk drive and EFS mounts as a network filesystem.
3. Concurrency: S3 can serve global users, EBS connects to one instance (AZ specific) and EFS connects to thousands of servers concurrently.
4. Performance & Cost: EBS delivers sub-millisecond latency for databases. S3 offers the lowest cost per gigabyte for bulk data. EFS provides flexible shared access with auto-scaling capacity.

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When you're designing a cloud system, use this simple decision rule:

1. Need a fast boot disk or host a database on a single server? Choose Amazon EBS for maximum performance and low latency.
2. Need a shared folder accessible by multiple web servers or containers at the same time? Choose Amazon EFS for elastic, concurrent file access.
3. Need to store millions of user uploads, backups, or media files accessible via API or web URLs? Choose Amazon S3 for low cost and infinite durability.

In fact, modern enterprise applications often use all three together: EBS for the database disk, EFS for shared user files, and S3 for long-term backups and public media static assets.

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To wrap up: understanding these access patterns allows you to architect cloud applications that are both cost-effective and highly performant. Thank you for your time today! I'd be happy to open up the floor to any questions. Have a great day and I will see you in the next one!